KCA UNIVERSITY

MAIN CAMPUS

SYSTEM REQUIREMENTS SPECIFICATION

Pharmacy POS Management System

for MedPlus Pharmacy

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the complete software requirements for the Pharmacy POS Management System. This system is designed to automate and streamline pharmacy operations at MedPlus Pharmacy, a community pharmacy located in Nairobi, Kenya.

The purpose of this document is to provide a detailed specification of the functional and non-functional requirements for the Pharmacy POS Management System. It serves as a reference for developers, testers, project managers, and stakeholders involved in the development and deployment of the system.

This SRS covers all aspects of the system including sales transaction processing, medicine inventory management with expiry date tracking, prescription record management, receipt generation, and comprehensive reporting capabilities.

1.2 Intended Audience and Reading Suggestions

This document is intended for the following audiences:

- Development Team: Software developers who will implement the system. They should read the entire document, with particular focus on Section 3 (System Features) and Section 4 (External Interface Requirements).
- Project Manager: Responsible for project planning and coordination. Should focus on Sections 1, 2, and 6 for scope and constraints understanding.
- Quality Assurance Team: Testers who will validate the system. Should thoroughly review Section 3 (System Features) to develop comprehensive test cases.
- Pharmacy Management (MedPlus): End users and system owners. Should review Sections 1, 2, and 3 to ensure requirements align with business needs.
- Supervisor and Academic Reviewers: For academic evaluation and approval of the project scope and requirements.

The recommended reading sequence is: start with Section 1 (Introduction) for overview, then Section 2 (Overall Description) for context, followed by Section 3 (System Features) for detailed requirements.

1.3 Project Scope

The Pharmacy POS Management System is a computerized solution designed to replace the manual and semi-automated systems currently used at MedPlus Pharmacy. The system will provide the following capabilities:

- Sales transaction processing for both prescription and over-the-counter (OTC) medicines
- Real-time medicine inventory management with automatic stock level updates
- Expiry date tracking with automated alerts for medicines approaching expiration
- Prescription record management and linking to sales transactions
- Receipt generation and printing
- Low stock alerts and notifications
- Comprehensive reporting (sales, inventory, expiry, profit analysis)
- User management with role-based access control (Admin, Pharmacist, Cashier)

The objectives of the system are to improve accuracy, speed, and reliability in pharmacy sales operations; reduce human errors in pricing and inventory tracking; ensure regulatory compliance with the Pharmacy and Poisons Board of Kenya requirements; minimize financial losses from expired stock; and support informed decision-making through accurate reporting.

The system will NOT include the following in this initial version: integration with external systems such as NHIF (National Hospital Insurance Fund) or supplier databases; controlled substance tracking features; multi-branch or chain pharmacy management capabilities; and online pharmacy or e-commerce functionality.

2. Overall Description

2.1 Product Perspective

The Pharmacy POS Management System is a new, self-contained software product designed specifically for MedPlus Pharmacy. It is not a follow-on member of an existing product family nor a replacement for a specific existing system, though it will replace the current manual record-keeping practices.

The system operates as a standalone application with a client-server architecture. Users interact with the system through web browsers (client layer), the application logic processes transactions on the server (application layer), and all persistent data is stored in a database (database layer).

The major components of the system include:

- Client Layer: Web browser interface for user interaction
- Application Layer: Django-based web application handling business logic
- Database Layer: MySQL/PostgreSQL database for data persistence
- External Hardware: Barcode scanner, receipt printer, and cash drawer

2.2 Product Features

The Pharmacy POS Management System will provide the following major features:

2.2.1 User Authentication and Authorization

Secure login system with role-based access control supporting three user roles: Administrator (full system access), Pharmacist (inventory and prescription management), and Cashier (sales processing and receipt generation).

2.2.2 Medicine Inventory Management

Complete inventory tracking including medicine master data, batch management, expiry date tracking, stock level monitoring, and automated alerts for low stock and expiring medicines.

2.2.3 Sales Management

Point-of-sale functionality including item selection via barcode or search, pricing with discount capabilities, multiple payment methods (Cash, M-Pesa, Card), receipt generation, and sales history tracking.

2.2.4 Prescription Management

Recording and management of prescription details including prescriber information, patient details, prescribed medicines, and linking prescriptions to sales transactions for regulatory compliance.

2.2.5 Reporting and Analytics

Comprehensive reporting capabilities including daily/weekly/monthly sales reports, inventory status reports, expiring medicines reports, low stock reports, profit margin analysis, and export to PDF/Excel formats.

2.3 User Classes and Characteristics

The system is designed for the following user classes:

2.3.1 Administrator

The Administrator is typically the pharmacy owner or IT specialist responsible for system configuration and management. Characteristics include: high technical expertise, full system access, responsible for user management, system configuration, and data backup. Frequency of use: daily for system monitoring and weekly for administrative tasks.

2.3.2 Pharmacist

The Pharmacist is a licensed professional responsible for medicine dispensing and prescription management. Characteristics include: pharmaceutical knowledge, moderate technical skills, manages inventory, processes prescriptions, and verifies medicine information. Frequency of use: continuous during working hours.

2.3.3 Cashier

The Cashier handles sales transactions and customer interactions. Characteristics include: basic technical skills, focuses on sales processing, receipt generation, and customer service. Frequency of use: continuous during working hours.

2.3.4 Patient/Customer (Indirect)

Patients are the end beneficiaries of the system through improved service quality, accurate dispensing, and faster transaction processing. They do not directly interact with the system.

2.4 Operating Environment

The Pharmacy POS Management System will operate in the following environment:

2.4.1 Hardware Platform

- Server: Minimum Intel Core i3 processor, 4GB RAM, 256GB storage (for development); recommended higher specifications for production
- Client Workstations: Standard desktop/laptop computers with web browser capability
- Barcode Scanner: USB handheld scanner for medicine identification
- Receipt Printer: 80mm thermal printer for sales receipts
- Cash Drawer: Electronic cash drawer connected to receipt printer

2.4.2 Software Environment

- Operating System: Windows 10/11 or Linux (Ubuntu recommended)
- Web Server: Django development server (development) or Apache/Nginx (production)
- Database: SQLite (development), MySQL 8.0+ or PostgreSQL 13+ (production)
- Web Browser: Chrome, Firefox, Edge (latest versions)
- Python: Version 3.9 or higher
- Django Framework: Version 4.x

2.5 Design and Implementation Constraints

The following constraints will affect the design and implementation of the system:

- Regulatory Compliance: The system must comply with Pharmacy and Poisons Board of Kenya regulations regarding prescription record keeping and medicine dispensing
- Academic Timeline: Development must be completed within the academic semester timeframe (approximately 6 weeks)
- Budget Limitations: Limited financial resources for hardware acquisition (estimated budget: KES 46,000 / USD 353)
- Technology Stack: Must use Python and Django framework as specified in the project proposal
- Open Source Requirement: Development tools and software must be open source to minimize costs
- Single Location: Initial version designed for single pharmacy location only
- Internet Dependency: Basic functionality should work with limited internet connectivity

2.6 Assumptions and Dependencies

The following assumptions and dependencies apply to this project:

2.6.1 Assumptions

- MedPlus Pharmacy staff will be available for requirements gathering and user acceptance testing
- Pharmacy management will provide necessary medicine data for initial system setup
- Required hardware (computer, barcode scanner, printer) will be available for deployment
- Users have basic computer literacy and can operate web-based applications
- Electricity supply is stable at the pharmacy location

2.6.2 Dependencies

- Django Framework: The system depends on Django 4.x for web application functionality
- Python Libraries: Dependencies include python-barcode for barcode generation, ReportLab/WeasyPrint for PDF generation, and Chart.js for visualization
- Database System: Requires MySQL or PostgreSQL for production deployment
- Browser Compatibility: System functionality depends on modern web browser capabilities
- Hardware Drivers: Proper functioning of barcode scanner and receipt printer depends on compatible drivers

3. System Features

3.1 User Authentication and Authorization Module

3.1.1 Description and Priority

This module manages user access to the system through secure login/logout functionality and implements role-based access control. Priority: HIGH. This is a critical security feature that must be implemented first.

3.1.2 Stimulus/Response

Stimulus: User enters username and password on login page. Response: System validates credentials and redirects to appropriate dashboard based on user role.

Stimulus: User clicks logout button. Response: System terminates session and redirects to login page.

Stimulus: User attempts to access unauthorized functionality. Response: System displays access denied message and logs the attempt.

3.1.3 Functional Requirements

REQ-AUTH-01: The system shall provide a login page accessible from the main application URL

REQ-AUTH-02: The system shall authenticate users based on username and password

REQ-AUTH-03: The system shall store passwords using secure hashing (bcrypt/SHA-256)

REQ-AUTH-04: The system shall support three user roles: Administrator, Pharmacist, and Cashier

REQ-AUTH-05: The system shall enforce role-based access control for all system functions

REQ-AUTH-06: The system shall maintain user session state during active use

REQ-AUTH-07: The system shall automatically log out inactive users after 30 minutes

REQ-AUTH-08: The system shall display appropriate error messages for invalid login attempts

REQ-AUTH-09: The system shall lock user accounts after 5 consecutive failed login attempts

REQ-AUTH-10: The system shall log all login/logout activities for audit purposes

3.2 Medicine Inventory Management Module

3.2.1 Description and Priority

This module manages all aspects of medicine inventory including master data, stock levels, batch tracking, and expiry date management. Priority: HIGH. Core functionality for pharmacy operations.

3.2.2 Stimulus/Response

Stimulus: Administrator adds new medicine to inventory. Response: System creates medicine record and updates inventory database.

Stimulus: Sale transaction completed. Response: System deducts sold quantity from inventory and updates stock levels.

Stimulus: Stock level falls below reorder point. Response: System generates low stock alert notification.

Stimulus: Medicine approaches expiry date (within 30 days). Response: System generates expiry alert notification.

3.2.3 Functional Requirements

REQ-INV-01: The system shall store medicine information including: Medicine ID/Barcode, Name (generic and brand), Category, Manufacturer, Unit price (buying and selling), Quantity in stock, Batch number, Expiry date, and Reorder level

REQ-INV-02: The system shall support medicine categories: Prescription, OTC, Supplements, and Medical Supplies

REQ-INV-03: The system shall allow adding new medicines to inventory

REQ-INV-04: The system shall allow updating existing medicine information

REQ-INV-05: The system shall allow deactivating (not deleting) discontinued medicines

REQ-INV-06: The system shall automatically update stock levels when sales are processed

REQ-INV-07: The system shall support batch management for medicines with different expiry dates

REQ-INV-08: The system shall implement FIFO (First-In-First-Out) for stock deduction

REQ-INV-09: The system shall generate low stock alerts when quantity falls below reorder level

REQ-INV-10: The system shall generate expiry alerts for medicines expiring within 30 days

REQ-INV-11: The system shall prevent dispensing of expired medicines

REQ-INV-12: The system shall support barcode scanning for medicine identification

REQ-INV-13: The system shall provide inventory search by name, category, or barcode

REQ-INV-14: The system shall display current stock status in real-time

3.3 Sales Management Module

3.3.1 Description and Priority

This module handles all sales transactions including item selection, pricing, payment processing, and receipt generation. Priority: HIGH. Primary function of the POS system.

3.3.2 Stimulus/Response

Stimulus: Cashier scans medicine barcode or selects from search. Response: System adds item to current transaction with price and updates running total.

Stimulus: Cashier applies discount. Response: System recalculates total with discount applied.

Stimulus: Cashier selects payment method and completes sale. Response: System records transaction, updates inventory, generates receipt, and opens cash drawer.

3.3.3 Functional Requirements

REQ-SALE-01: The system shall support item selection via barcode scanning

REQ-SALE-02: The system shall support item selection via search by name

REQ-SALE-03: The system shall display item details including name, price, and availability

REQ-SALE-04: The system shall calculate subtotal, tax (if applicable), and total amount automatically

REQ-SALE-05: The system shall support percentage and fixed amount discounts

REQ-SALE-06: The system shall support multiple payment methods: Cash, M-Pesa, and Card

REQ-SALE-07: The system shall calculate and display change for cash payments

REQ-SALE-08: The system shall generate and print receipts with pharmacy details, transaction information, and items sold

REQ-SALE-09: The system shall open the cash drawer automatically upon cash payment completion

REQ-SALE-10: The system shall record each transaction with unique ID, date/time, items, amounts, and cashier information

REQ-SALE-11: The system shall allow voiding transactions with appropriate authorization

REQ-SALE-12: The system shall support prescription sales with mandatory prescription linking

REQ-SALE-13: The system shall provide sales history lookup by date, transaction ID, or customer

REQ-SALE-14: The system shall prevent sale of expired or out-of-stock medicines

3.4 Prescription Management Module

3.4.1 Description and Priority

This module manages prescription records and links them to sales transactions for regulatory compliance. Priority: HIGH. Required for legal compliance with pharmacy regulations.

3.4.2 Stimulus/Response

Stimulus: Pharmacist enters prescription details. Response: System creates prescription record and associates it with the sale transaction.

Stimulus: User searches for prescription history. Response: System displays prescription records with linked transactions.

3.4.3 Functional Requirements

REQ-PRES-01: The system shall record prescription details including: Prescription ID, Patient name, Prescriber name and facility, Date issued, and Medicines prescribed

REQ-PRES-02: The system shall link prescriptions to sales transactions

REQ-PRES-03: The system shall maintain prescription history for compliance and audit purposes

REQ-PRES-04: The system shall require prescription information for prescription medicine sales

REQ-PRES-05: The system shall allow searching prescriptions by patient name, prescriber, or date

REQ-PRES-06: The system shall display prescription details on receipts for prescription sales

REQ-PRES-07: The system shall prevent modification of prescription records after completion

3.5 Reporting Module

3.5.1 Description and Priority

This module generates various reports for management decision-making and operational monitoring. Priority: MEDIUM. Important for business intelligence but not critical for daily operations.

3.5.2 Stimulus/Response

Stimulus: User selects report type and date range. Response: System generates and displays the requested report.

Stimulus: User clicks export button. Response: System exports report to PDF or Excel format.

3.5.3 Functional Requirements

REQ-REP-01: The system shall generate daily sales reports showing transactions, revenue, and items sold

REQ-REP-02: The system shall generate weekly sales reports with comparative analysis

REQ-REP-03: The system shall generate monthly sales reports with trends and summaries

REQ-REP-04: The system shall generate inventory status reports showing current stock levels

REQ-REP-05: The system shall generate expiring medicines reports with date range filtering

REQ-REP-06: The system shall generate low stock reports for reorder planning

REQ-REP-07: The system shall generate profit margin analysis reports

REQ-REP-08: The system shall support date range filtering for all reports

REQ-REP-09: The system shall export reports to PDF format

REQ-REP-10: The system shall export reports to Excel format

REQ-REP-11: The system shall display reports with interactive charts using Chart.js

REQ-REP-12: The system shall allow printing of generated reports

3.6 User Management Module

3.6.1 Description and Priority

This module manages system users, their roles, and activity monitoring. Priority: MEDIUM. Administrative function for system security.

3.6.2 Stimulus/Response

Stimulus: Administrator adds new user. Response: System creates user account with specified role.

Stimulus: Administrator deactivates user. Response: System disables account access while preserving records.

3.6.3 Functional Requirements

REQ-USER-01: The system shall allow Administrators to add new users

REQ-USER-02: The system shall allow Administrators to update user information

REQ-USER-03: The system shall allow Administrators to deactivate users (soft delete)

REQ-USER-04: The system shall allow Administrators to assign roles to users

REQ-USER-05: The system shall store user information including: User ID, Username, Password (hashed), Full name, Role, Contact information, and Status

REQ-USER-06: The system shall maintain user activity logs

REQ-USER-07: The system shall allow users to change their own passwords

REQ-USER-08: The system shall enforce unique usernames

REQ-USER-09: The system shall display user activity logs to Administrators

4. External Interface Requirements

4.1 User Interfaces

The system will provide a web-based user interface accessible through standard web browsers. The interface design follows these principles:

- Responsive Design: Interface adapts to different screen sizes (desktop/laptop focus)
- Intuitive Navigation: Clear menu structure with role-based menu visibility
- Consistent Layout: Standard header, sidebar navigation, and content area across all pages
- Color Coding: Different colors for different medicine categories and alert types
- Error Messages: Clear, user-friendly error messages in English

4.1.1 Login Interface

Simple login form with username and password fields, login button, and error message display area.

4.1.2 Dashboard Interface

Role-specific dashboard displaying relevant information: quick actions, alerts, summary statistics, and navigation menu.

4.1.3 Sales Interface

Point-of-sale screen with item entry area (barcode/search), cart display, pricing information, payment method selection, and receipt preview.

4.1.4 Inventory Interface

Medicine listing with search and filter options, add/edit forms, stock level indicators, and expiry date alerts.

4.1.5 Reporting Interface

Report selection menu, date range picker, report display area with charts, and export buttons.

4.2 Hardware Interfaces

The system interfaces with the following hardware components:

4.2.1 Barcode Scanner

- Interface: USB HID (Human Interface Device)
- Data Format: ASCII text representing barcode numbers
- Integration: Scanner input captured as keyboard input in barcode fields
- Supported Types: EAN-13, UPC-A, Code 128

4.2.2 Receipt Printer

- Interface: USB or Ethernet connection
- Type: 80mm thermal printer
- Integration: Print commands via printer driver
- Features: Auto-cut, cash drawer trigger signal

4.2.3 Cash Drawer

- Interface: RJ11/RJ12 connection to receipt printer
- Integration: Triggered by printer kick command
- Operation: Opens automatically after cash sale completion

4.3 Software Interfaces

The system interfaces with the following software components:

4.3.1 Database Management System

- Component: MySQL 8.0+ or PostgreSQL 13+
- Interface: Django ORM (Object-Relational Mapping)
- Data: All persistent application data

4.3.2 Web Server

- Development: Django development server
- Production: Apache with mod_wsgi or Nginx with Gunicorn
- Protocol: HTTP/HTTPS

4.3.3 External Libraries

- python-barcode: Barcode generation (version 0.14.0+)
- ReportLab/WeasyPrint: PDF generation for receipts and reports
- Chart.js: Client-side chart rendering (version 3.x+)
- Bootstrap 5: Frontend UI framework

4.4 Communications Interfaces

The system uses the following communication protocols and interfaces:

4.4.1 Network Protocols

- HTTP/HTTPS: Web application communication
- TCP/IP: Underlying network protocol

4.4.2 Data Formats

- JSON: Data exchange between frontend and backend
- HTML/CSS: Web page rendering
- JavaScript: Client-side scripting
- PDF: Report and receipt export format
- CSV/Excel: Data export format

4.4.3 Security

- HTTPS: Encrypted communication for production deployment

- Session-based authentication: Secure session management
- CSRF protection: Django built-in CSRF token validation

5. Nonfunctional Requirements

5.1 Performance Requirements

REQ-PERF-01: The system shall respond to user actions within 2 seconds under normal load

REQ-PERF-02: The system shall support at least 5 concurrent users without performance degradation

REQ-PERF-03: The system shall process sales transactions within 3 seconds from submission to receipt generation

REQ-PERF-04: The system shall generate reports within 10 seconds for date ranges up to 30 days

REQ-PERF-05: The system shall support inventory of up to 10,000 medicine items

REQ-PERF-06: The system shall maintain transaction history for at least 2 years

REQ-PERF-07: The system shall have 99% uptime during pharmacy operating hours

5.2 Safety Requirements

REQ-SAFE-01: The system shall prevent dispensing of expired medicines with clear warnings

REQ-SAFE-02: The system shall require prescription verification for prescription-only medicines

REQ-SAFE-03: The system shall maintain accurate pricing to prevent customer overcharging

REQ-SAFE-04: The system shall provide confirmation prompts for critical actions (void, delete)

REQ-SAFE-05: The system shall maintain data integrity through regular automated backups

REQ-SAFE-06: The system shall comply with Pharmacy and Poisons Board of Kenya regulations

5.3 Security Requirements

REQ-SEC-01: The system shall require authentication for all access

REQ-SEC-02: The system shall store passwords using secure one-way hashing

REQ-SEC-03: The system shall implement role-based access control

REQ-SEC-04: The system shall encrypt sensitive data in transit using HTTPS

REQ-SEC-05: The system shall log all security-relevant events (login, logout, access violations)

REQ-SEC-06: The system shall implement session timeout after 30 minutes of inactivity

REQ-SEC-07: The system shall protect against SQL injection through parameterized queries

REQ-SEC-08: The system shall protect against cross-site scripting (XSS) attacks

REQ-SEC-09: The system shall enforce account lockout after 5 failed login attempts

REQ-SEC-10: The system shall require strong passwords (minimum 8 characters with mixed case, numbers)

6. Other Requirements

6.1 Database Requirements

- The database shall support ACID transactions for data integrity
- The database schema shall be normalized to at least 3rd Normal Form (3NF)
- The database shall support foreign key constraints for referential integrity
- The database shall implement appropriate indexing for query performance

6.2 Legal and Regulatory Requirements

- The system shall comply with Pharmacy and Poisons Board of Kenya regulations
- The system shall maintain prescription records as required by law
- The system shall comply with data protection regulations for patient information

6.3 Documentation Requirements

- User Manual: Comprehensive guide for pharmacy staff
- Technical Documentation: System architecture and code documentation
- Training Materials: Quick reference guides and training presentations

6.4 Portability Requirements

- The system shall be deployable on standard Windows or Linux servers
- The system shall be accessible from any modern web browser
- The system database shall be portable between MySQL and PostgreSQL

Appendix A: Glossary

ACID: Atomicity, Consistency, Isolation, Durability - properties of database transactions

API: Application Programming Interface - set of protocols for building software applications

Barcode: Machine-readable representation of data in visual form

Cashier: System user role responsible for processing sales transactions

CRUD: Create, Read, Update, Delete - basic database operations

CSRF: Cross-Site Request Forgery - type of security vulnerability

FIFO: First-In-First-Out - inventory management method

M-Pesa: Mobile money payment service widely used in Kenya

NHIF: National Hospital Insurance Fund - Kenyan health insurance scheme

ORM: Object-Relational Mapping - technique for converting data between systems

OTC: Over-The-Counter - medicines that can be sold without prescription

PDF: Portable Document Format - file format for documents

Pharmacist: Licensed professional authorized to dispense prescription medicines

POS: Point of Sale - location and system where transactions occur

PPB: Pharmacy and Poisons Board - Kenyan regulatory body for pharmacies

Prescription: Medical instruction for dispensing medicine to a patient

SRS: Software Requirements Specification - this document

SSL/TLS: Secure Sockets Layer/Transport Layer Security - encryption protocols

UAT: User Acceptance Testing - validation by end users

UI: User Interface - visual elements users interact with

URL: Uniform Resource Locator - web address

XSS: Cross-Site Scripting - type of security vulnerability

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